

BEST FIT

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manishaa@localhost:~$ # BEST FIT COMMENTS - PART 1

# 1. General Purpose
# This script demonstrates good commenting practices.
# Comments improve code readability and maintainability.
# Always comment what the code does, not how.

# 2. Variables
# Store user information
name="Manishaa"      # User's name
age=22               # User's age
city="Hyderabad"     # User's city

# 3. Function to Greet User
# This function prints a personalized greeting message.
greet() {
    echo "Hello, $name! Welcome to Linux shell scripting."
}

# 4. Function to Add Two Numbers
# Takes two numbers as arguments and returns their sum.
add() {
    sum=$(( $1 + $2 )) # Perform arithmetic addition
    echo "Sum is: $sum" # Display the result
}

# 5. Main Logic
# Call functions and display output
greet                # Call greet function
add 10 20            # Call add function with arguments
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manishaa@localhost:~$ # BEST FIT COMMENTS - PART 2

# 1. Conditional Statement
# Check if the user is an adult or minor.
if [ $age -ge 18 ]; then # If age is greater than or equal to 18
    echo "You are an adult." # Execute if condition is true
else
    echo "You are a minor." # Execute if condition is false
fi # End of if-else block

# 2. For Loop Example
# Print numbers from 1 to 5 using a for loop.
for i in {1..5}; do # Start loop from 1 to 5
    echo "Number: $i" # Print current number
done # End of loop

# 3. While Loop Example
# Keep running while count is less than or equal to 3.
count=1 # Initialize counter
while [ $count -le 3 ]; do # Loop until count <= 3
    echo "Count: $count" # Print count
    ((count++)) # Increment counter
done # End of while loop
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manishaa@localhost:~$ # BEST FIT COMMENTS - PART 3

# 1. File Operations
# Check if a file exists before reading.
file="sample.txt" # File name
if [ -f $file ]; then # If file exists
    echo "File exists." # Inform user
    cat $file # Display file content
else
    echo "File does not exist." # If file not found
fi # End of file check

# 2. Backup a File
# Create a backup of the file with timestamp.
if [ -f $file ]; then
    cp $file ${file}.${date +%Y%m%d%H%M%S}.bak # Copy file with timestamp
    echo "Backup created successfully." # Backup success message
fi

# 3. Remove Temporary Files
# Delete all .tmp files in the current directory.
rm -f *.tmp # Force remove temporary files
echo "Temporary files cleaned." # Confirmation message
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manishaa@localhost:~$ # BEST FIT COMMENTS - PART 4

# 1. User Input Validation
# Ask user for input and validate it.
read -p "Enter your age: " user_age # Read age from user
if [[ $user_age =~ ^[0-9]+$ ]]; then # Check if input is a number
    echo "You entered: $user_age" # Valid input
else
    echo "Invalid input! Please enter a number." # Invalid input
fi

# 2. Function with Return Value
# Returns the square of a number.
square() {
    local num=$1 # Local variable
    echo $(( num * num )) # Return square
}

# 3. Call Function and Store Result
result=$(square 7) # Call function and store output
echo "Square is: $result" # Display result

# 4. End of Script
# Thank the user and exit gracefully.
echo "Thank you, $name! Keep learning and keep growing."
exit 0 # Exit with status 0 (success)
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